

Healthcare Capstone: Diabetes Risk Prediction

4-Model Comparative Analysis & Clinical Web Deployment

Dataset: Pima Indians Diabetes Diagnostic Database (768 Patients, 8 Diagnostic Biomarkers)

Team Roles: Member 1 (M1 Baseline) | Member 2 (M2 Polynomial) | Member 3 (M3 Tree) | Member 4 (M4 Ensemble & Web App)

1. Problem Statement & Clinical Data Overview

- Objective: Build an early diagnostic machine learning screening tool to classify diabetes risk.
- Clinical Data: 768 female patient observations with 8 physiological measurements and binary Outcome.
- Medical Zero Detection: Glucose, Blood Pressure, Skin Thickness, Insulin, and BMI had impossible 0 values.
- Imputation Strategy: Replaced 0s with training-set medians strictly to prevent data leakage.
- Feature Engineering: Added WHO BMI Categories (0-3) and Glucose-to-Insulin Metabolic Ratio.

2. Four Machine Learning Architectures

- M1: Logistic Regression - Linear baseline model with Standard Scaling and L2 regularization ($C=1.0$).
- M2: Polynomial Logistic ($\text{deg}=2$) - Expands 10 features to 65 interaction terms (e.g. Glucose x BMI) with $C=0.1$.
- M3: Decision Tree Classifier - Pruned with 5-fold GridSearchCV ($\text{max_depth}=8$, $\text{min_samples_split}=20$).
- M4: Random Forest Classifier - Ensemble of 150 decorrelated decision trees with feature sub-sampling (sqrt).

3. Performance Comparison on 80:20 Unseen Test Split

	Model Architecture	Test Accuracy	Precision	Recall	F1-Score	5-Fold CV Accuracy
M1: Logistic Regression		69.48%	0.5778	0.4815	0.5253	78.82% (+-1.33%)
	M2: Polynomial Logistic (degree 2)	73.38%	0.6512	0.5185	0.5773	78.50% (+-1.36%)
	M3: Decision Tree (Tuned)	74.03%	0.6667	0.5185	0.5833	71.82% (+-2.93%)
M4: Random Forest (Tuned)		75.97%	0.6889	0.5741	0.6263	77.85% (+-1.42%)

4. Winning Model Justification & Clinical Safety

- Undisputed Winner: Random Forest achieved #1 in Test Accuracy (75.97%), Recall (57.41%), and F1 (0.6263).
- Medical Priority (Recall): Catching real diabetic patients minimizes dangerous undiagnosed complications.
- Cross-Validation Stability: Maintained 77.85% across all 5 folds without overfitting (unlike Decision Tree).
- Confusion Matrix: 86 True Negatives, 31 True Positives, with only 14 False Alarms on unseen test patients.
- Top Predictive Biomarkers: Glucose (33.3%), BMI (13.4%), and Age (11.9%) drive clinical diagnosis.

5. Streamlit Web Deployment & System Features

- Interactive Diagnostic Tool: Real-time risk probability output (High Risk Diabetic vs Low Risk Healthy).
- Dynamic Imputation: User can enter clinical vitals; unknown fields safely default to training medians.
- Multi-Model Switching: Clinicians can toggle between all 4 trained models in the sidebar.
- Visual Analytics Tabs: Built-in tabs displaying EDA plots, Feature Importances, and Benchmark Tables.
- Local Serving: Executed with a single command: `streamlit run app.py`